

AUGUST/RESIT EXAMINATIONS 2023/2024

MODULE: CA216 - Operating Systems

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing)
CASE	BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 2,0

EXAMINER(S):

Dr. Graham Healy	(Internal)	(Ext:7364)
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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]****Q 1(a)****[6 Marks]**

Explain the differences between a process and a thread. Provide an example of a situation where using a process is appropriate and another example where using a thread is appropriate.

Q 1(b)**[11 Marks]**

- a) Explain the purpose of the `fork()` system call and describe what happens in both the parent and child processes after a `fork()` call.
- b) Explain the purpose of the `exec()` system call and describe how it changes the behavior of the process that calls it.
- c) Provide a scenario where using both `fork()` and `exec()` are used together. Explain the sequence of calls and the expected outcomes.

Q 1(c)**[4 Marks]**

What is a dynamically linked library? What is a benefit of using a dynamically linked library?

Q 1(d)**[4 Marks]**

Briefly explain what the `ps tree` command does on Linux. In your answer explain the relationship it shows between processes.

[End of Question 1]**QUESTION 2****[TOTAL MARKS: 25]****Q 2(a)****[5 Marks]**

Using Amdahl's law, calculate the speedup of a program with a 50% serial proportion for 10 processors.

Q 2(b)**[6 Marks]**

The `kill` command on Linux can be used to send a signal to a process. Explain what a signal is and explain why it can be beneficial for a process to receive signals.

Q 2(c)**[6 Marks]**

Consider the following C program that will be compiled and executed on Linux. The writer of the code expected an output of 100,000 to be printed for both the 1st and 2nd counter i.e. on lines 22 and 23. While the value 100,000 was printed for the 2nd counter, the value 66,149 was printed for the 1st counter. With reference to specific line numbers, briefly explain why there is a difference in the printed values.

```
01: #include <stdio.h>
02: #include <threads.h>
03: #include <stdatomic.h>

04: atomic_int acnt;
05: int cnt;

06: int f(void* thr_data)
07: {
08:     for(int n = 0; n < 10000; ++n)
09:     {
10:         ++cnt;
11:         ++acnt;
12:     }

13:     return 0;
14: }

15: int main(void)
16: {
17:     thrd_t thr[10];
18:     for(int n = 0; n < 10; ++n)
19:         thrd_create(&thr[n], f, NULL);
20:     for(int n = 0; n < 10; ++n)
21:         thrd_join(thr[n], NULL);

22:     printf("The 1st counter is %d \n", cnt);
23:     printf("The 2nd counter is %d \n", acnt);
24: }
```

Q 2(d)**[8 Marks]**

Priority scheduling with round robin is used to schedule the processes in the table below for execution on a single-core CPU with a time quantum of 2. Using a Gantt Chart, illustrate the execution order of the processes.

Process	Priority	Burst Time
P1	3	4
P2	2	5
P3	2	8
P4	1	7
P5	3	3

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[12 Marks]**

Explain the difference between deadlock prevention and deadlock avoidance. In your answer describe the necessary conditions for deadlock to occur.

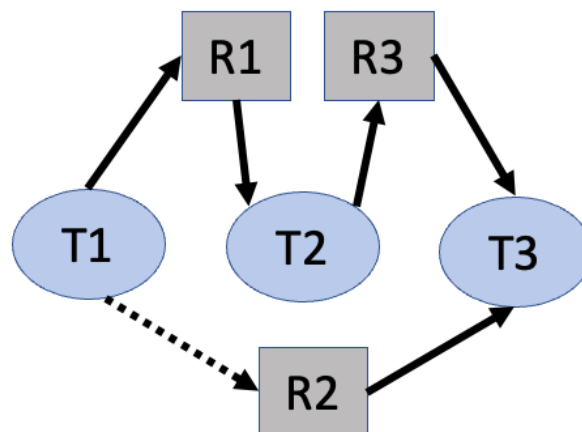
Q 3(b)**[7 Marks]**

Explain what busy waiting / spinlock is. Why is it not advantageous to use busy waiting / spinlock? What alternatives could be used instead of busy waiting / spinlock?

Q 3(c)**[6 Marks]**

Consider the resource-allocation graph below for a system where a single instance is available for each resource (i.e. R1, R2, and R3). T1/T2/T3 are tasks/processes. Is this system in a safe, unsafe or deadlocked state? Explain your answer.

Note: a broken-arrow line indicates a claim edge.



[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[5 Marks]**

The *chmod* command is used to set file permissions. Explain what file permissions are set for *filename* with the following command:

chmod 761 filename

Q 4(b)

[4 Marks]

Explain the difference between hard and soft links in the context of a filesystem.

Q 4(c)

[11 Marks]

Paging is a memory management scheme. Explain how paging works using an example of a 32-bit virtual address. In your answer, explain how *swapping* can be used as part of paging memory management.

Q 4(d)

[5 Marks]

Explain what is meant by the term DMA (Direct Memory Access).

[End of Question 4]

[END OF EXAM]